

COMPUTER VISION

Neural Network Miscellaneous

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BACKPROPAGATION

LOSS FUNCTION

OPTIMIZER

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FRAMEWORK EXAMPLES

WHAT IS A NEURAL NETWORK?

- Neural networks are function approximators.
- Composed of layers: input, hidden, output.
- Learning by adjusting weights.

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BACKPROPAGATION OVERVIEW

- Backpropagation: method to compute gradients.
- Chain rule of calculus applied to the loss function.

MATHEMATICS OF BACKPROPAGATION

Let:

- L be the loss function.
- w be the weights.
- a be the activations.

The update rule is:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$

where η is the learning rate.

Gradients are computed via:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \frac{\partial a}{\partial w}$$

BACKPROPAGATION ALGORITHM

1. Forward pass: compute outputs.
2. Compute loss.
3. Backward pass: compute gradients via chain rule.
4. Update parameters.

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LOSS FUNCTION

- Measures the difference between predicted and true values.
- Guides the learning process.

EXAMPLES OF LOSS FUNCTIONS

Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Cross-Entropy Loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^n y_i \log(\hat{y}_i)$$

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WHAT IS AN OPTIMIZER?

- Algorithm that updates the model parameters.
- Based on gradients calculated from the loss.

STOCHASTIC GRADIENT DESCENT (SGD)

Update rule:

$$w_{t+1} = w_t - \eta \nabla L(w_t)$$

ADAM OPTIMIZER

Adam combines momentum and adaptive learning rate:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla L(w_t)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla L(w_t))^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

$$w_{t+1} = w_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

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LEARNING RATE

- Controls the size of weight updates.
- Critical for convergence speed and stability.

IMPACT OF LEARNING RATE

- **Too large:** May cause divergence.
- **Too small:** Slow convergence, stuck at local minima.

LEARNING RATE SCHEDULING

Common techniques:

- Step decay
- Exponential decay
- Cosine annealing

Dynamic adjustment can help training efficiency.

EXAMPLE OF LEARNING RATE DECAY

Exponential Decay:

$$\eta_t = \eta_0 e^{-\lambda t}$$

where:

- η_t : learning rate at epoch t
- η_0 : initial learning rate
- λ : decay rate

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EPOCH AND BATCH SIZE

- **Epoch:** One pass through the entire dataset.
- **Batch Size:** Number of samples per gradient update.

IMPACT OF BATCH SIZE

- Small batch: noisy gradient, faster updates.
- Large batch: stable gradient, slower updates.

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SUMMARY

- Backpropagation: Compute gradients.
- Loss Function: Guide learning.
- Optimizer: Update parameters.
- Learning Rate: Control step size.
- Epoch and Batch Size: Structure the training process.

